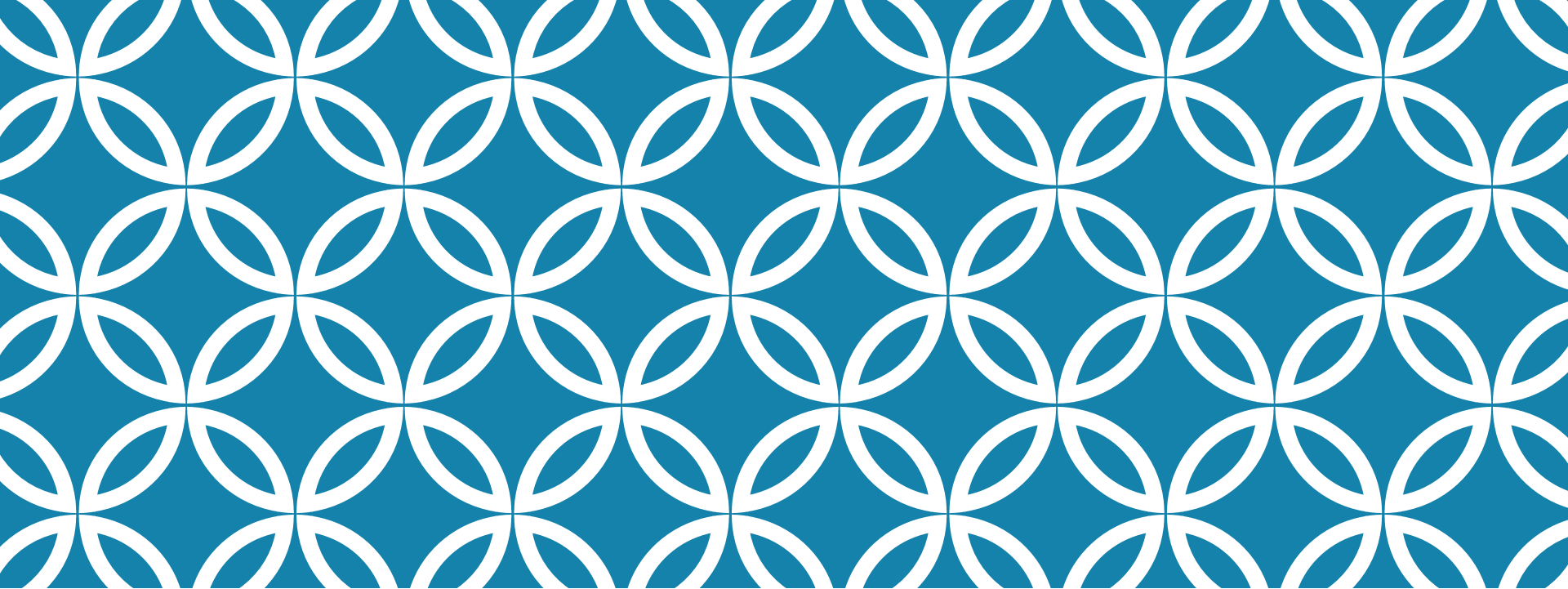




自動化驗收測試

第一階段(大小網)

目標

- ❖ 加速驗收測試
 - 同樣的功能測試應用在多個 site

測試環境(1/2)

❖ QA lab

Type	IP	Account	Password
vSphere Client	ALL	████	████████████████
遠端桌面	████████████	████████████	████████
遠端桌面	████████████	████████	████████

測試環境(2/2)

❖ 本機(VM player, VMware workstation, etc)

➤ OS: win 7, win 10

➤ 安裝Python3.6.3

➤ 開啟cmd 後執行

- pip install robotframework

- pip install future

- pip install -U

<https://github.com/HelioGuilherme66/robotframework-selenium2library/archive/v1.8.0b1.tar.gz>



測試工具



Robot Framework

- 驗收測試(python + selenium)
- 使用關鍵字, 不熟悉程式語言也能輕易上手
- 撰寫速度快
- 支援報表

編寫測試案例(1/6)

❖ 文件結構

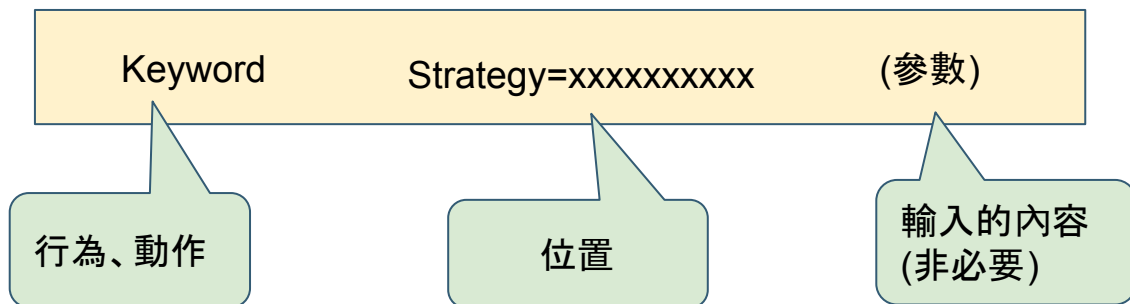
```
*** Variables ***
| ${BROWSER} | chrome
| ${URL} | 
| ${CREDENTIAL} | 
| ${WindowTitle};

*** Settings ***
| Library | Selenium2Library
| Suite Teardown | Close all browsers

*** Test Cases ***
| 打開瀏覽器
| | Open browser | ${URL} | ${BROWSER}
| | Maximize Browser Window
| 點選登入
| | Click Element | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/div/header/div[1]/div/div/div[2]/div/span[1]
| | Wait Until Element Is Visible | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/article/div/h1
| | Element Should Contain | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/article/div/h1 | 用戶登陸
| 輸入帳號登入
| | Input Text | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/article/div/form/div[1]/div/input | ${CREDENTIAL}
| | Input Text | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/article/div/form/div[2]/div/input | ${CREDENTIAL}
| | Click Element | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/article/div/form/div[3]/button
| | Wait Until Element Is Visible | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/div/header/div[1]/div/div/div[2]/div/div[1]/div[1]/span
| | Element Text Should Be | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/div/header/div[1]/div/div/div[2]/div/div[1]/div[1]/span | ${CREDENTIAL}
| 購彩大廳 快3 立即投注 全選
| | Click Element | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/div/header/div[2]/nav/ul/li[2]
| | Wait Until Element Is Visible | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/div/div/div/ul/li[5]/p
| | Click Element | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/div/div/div/ul/li[5]/p
| | Wait Until Element Is Visible | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/div/div/div/div/div[1]/div
| | Click Element | xpath=//*[@id="react-view"]/div/div/div[2]/div/div/div/div/div/div/div[1]/div/div[2]/button

*** Keywords ***
Scroll Page To Location
    [Arguments]    ${x_location}    ${y_location}
    Execute JavaScript    window.scrollTo(${x_location},${y_location})
```

編寫測試案例(2/6)



編寫測試案例(3/6)



Keyword

- Open browser
- Maximize Browser Window
- Click Element
- Wait Until Element Is Visible
- Element Should Contain
- Input Text
- Element Text Should Be
- Select Frame
- Select Window
- Get Title
- 其他可參考文件
- 自行定義: 如Scroll Page To Location

編寫測試案例(4/6)



Strategy

- css
- xpath
- id
- name
- link
- partial link
- tag

編寫測試案例(5/6)

❖ Strategy(以XPath為例)

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編寫測試案例(6/6)

- ❖ 當無法定位元素時，注意Window與iFrame的切換



執行測試案例

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觀看測試結果

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分配



題目:

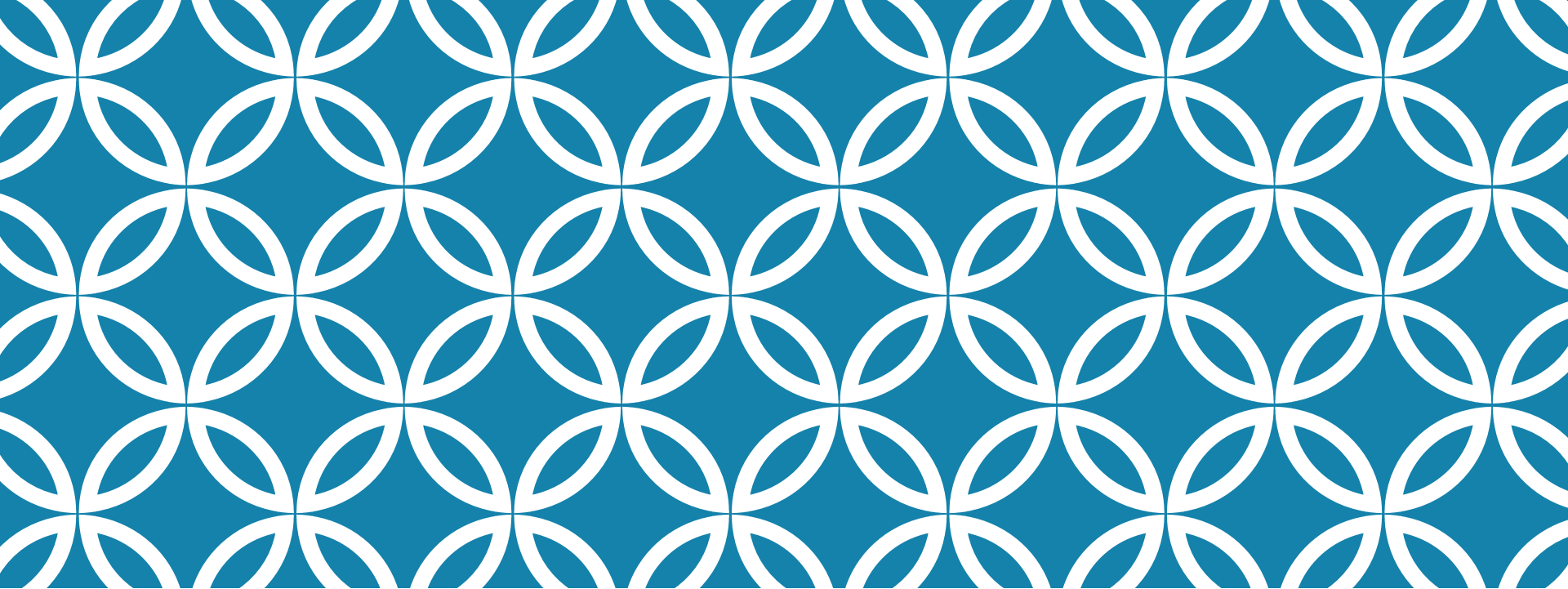
- 平台: 註冊、充值、提線、投注、撤單、每日加獎
- 遊戲: 大網選一個彩種遊戲所有按鈕全選 > 投注成功
- 繳交時間: 11/17(五)



方式: 將.robot檔案放到QA lab

Editor

<https://github.com/vkosuri/npp-robot>



自動化驗收測試

第二階段(APP)

Android 自動化測試

❖ 安裝環境

- Appium
- Android Studio 與 SDK
- Appium library for robot framework
- APP-inspector

❖ 取得APP元素定位

- 連接手機
- 啟動APP inspector

❖ 執行測試案例

- 啟動Appium
- 執行.robot

安裝Appium

- ❖ 下載並安裝nodejs
 - 檢查nodejs版本(開啟命令提示字元)
 - 輸入node -v
 - 輸入npm -v
- ❖ 安裝appium套件(以系統管理員身分開啟命令提示字元)
 - 輸入npm install -g appium
 - 輸入npm install -g appium-doctor

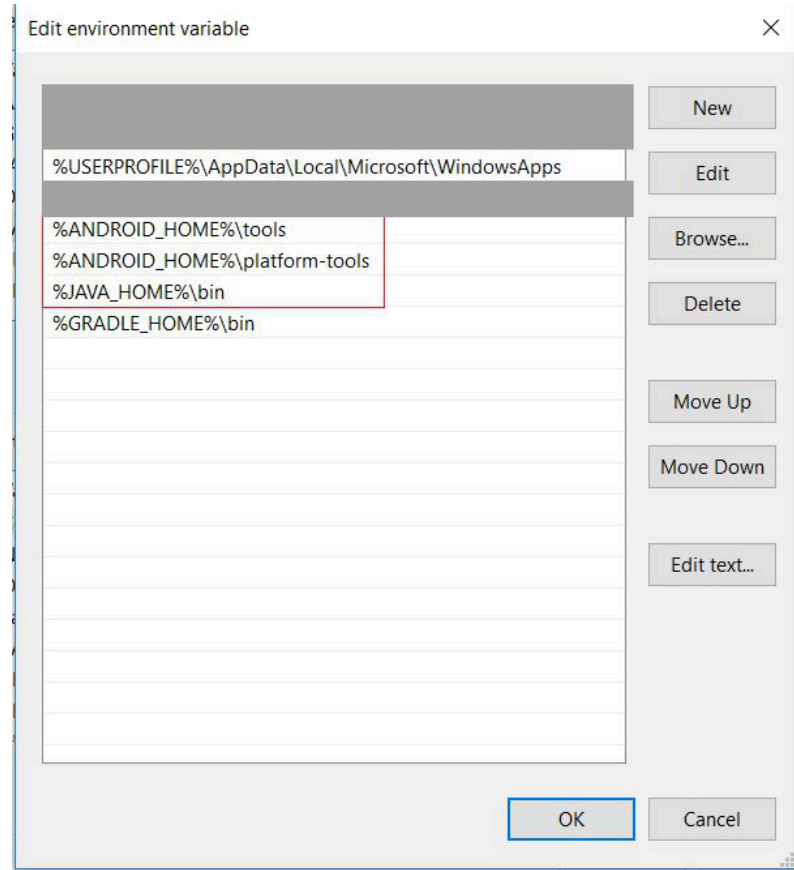
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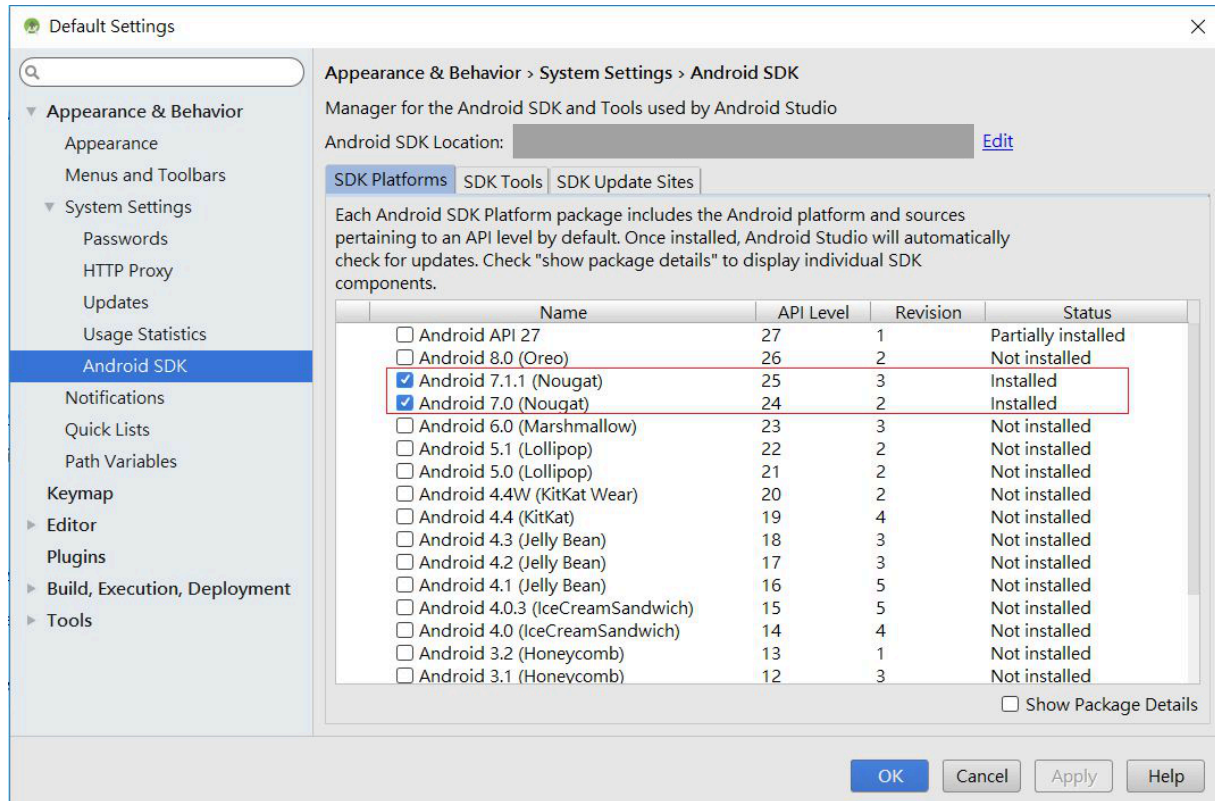
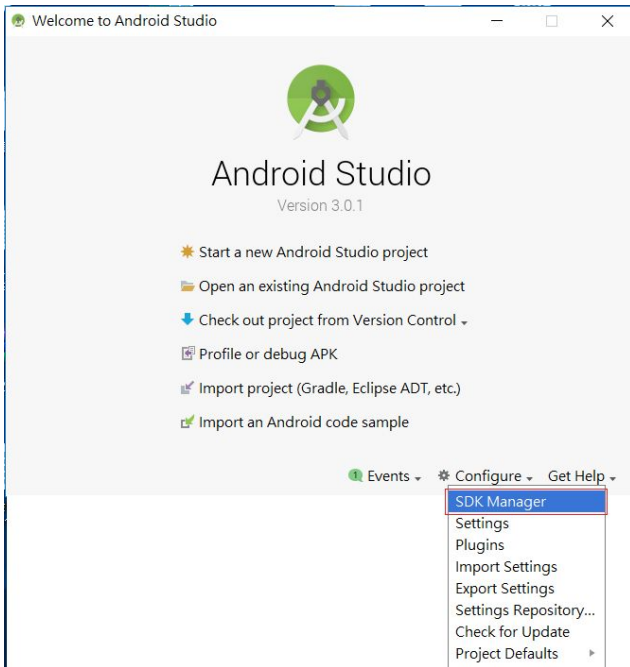
安裝Android studio與SDK

- ❖ 下載並安裝[Android Studio](#)
- ❖ 加入環境變數(如下表)
- ❖ 打開Android Studio > SDK manager > 勾選對應的Android版本

環境變數	路徑
ANDROID_HOME	C:\Users\<user_name>\AppData\Local\Android\Sdk
JAVA_HOME	C:\Program Files\Android\Android Studio\jre
path	%ANDROID_HOME%\tools %ANDROID_HOME%\platform-tools %JAVA_HOME%\bin

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檢查安裝情況

- ❖ 檢查安裝情況(開啟命令提示字元輸入appium-doctor)

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安裝Appium library for robot framework

- ❖ 在原本已安裝Robot Framework的環境下，開啟命令提示字元
 - `pip install robotframework-appiumlibrary`
 - `pip install six`

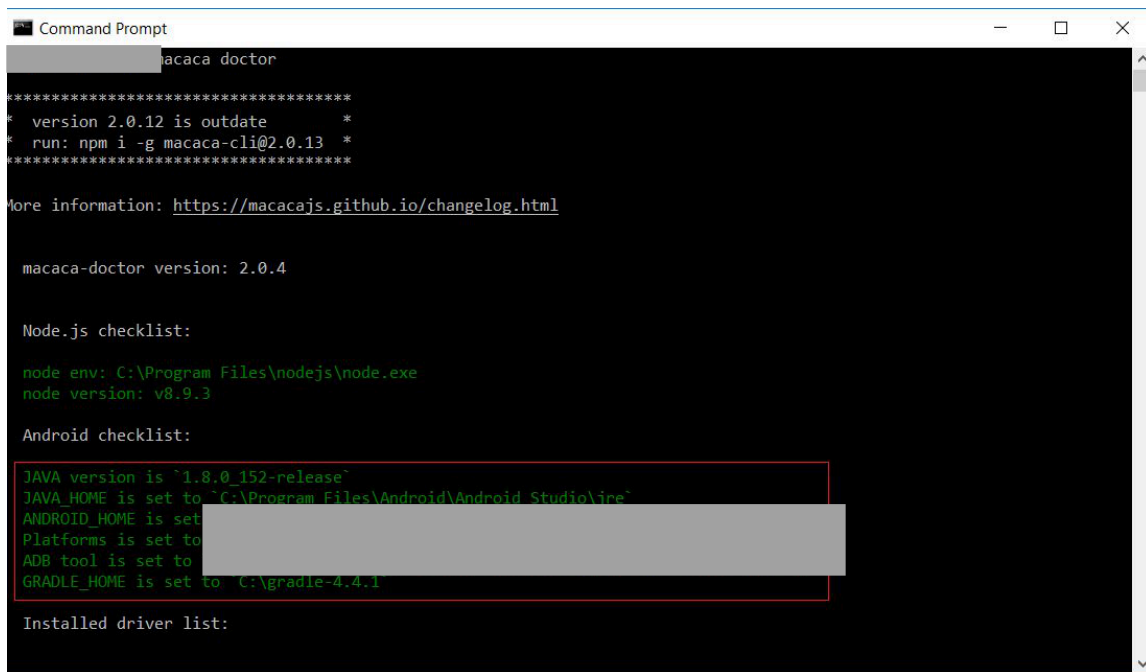
安裝App Inspector

- ❖ 安裝Macaca(開啟命令提示字元)
 - 輸入 `npm install macaca-cli -g`
- ❖ 安裝Gradle
 - 下載後解壓縮到指定目錄(ex. [REDACTED])
 - 新增環境變數(如下表)

環境變數	路徑
GRADLE_HOME	[REDACTED]
path	%GRADLE_HOME%\bin

檢查安裝情況

- ❖ 檢查安裝情況(開啟命令提示字元輸入macaca doctor)



```
Command Prompt
macaca doctor

*****
* version 2.0.12 is outdate      *
* run: npm i -g macaca-cli@2.0.13 *
*****

More information: https://macacajs.github.io/changelog.html

macaca-doctor version: 2.0.4

Node.js checklist:

node env: C:\Program Files\nodejs\node.exe
node version: v8.9.3

Android checklist:

JAVA version is `1.8.0_152-release`
JAVA_HOME is set to `C:\Program Files\Android\Android Studio\jre`
ANDROID_HOME is set to [REDACTED]
Platforms is set to [REDACTED]
ADB tool is set to [REDACTED]
GRADLE_HOME is set to C:\gradle-4.4.1

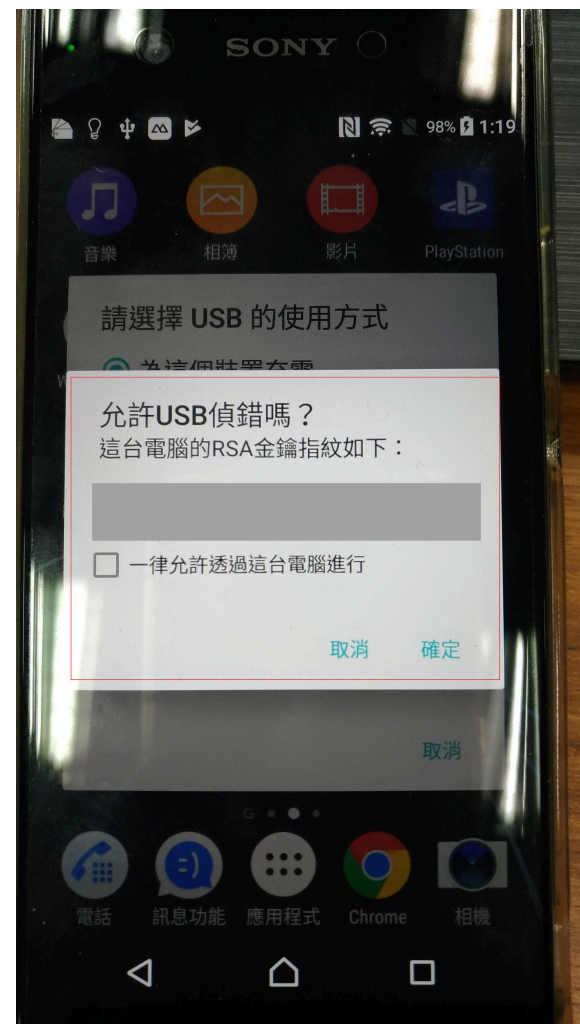
Installed driver list:
```

安裝App Inspector

- ❖ 安裝App Inspector(開啟命令提示字元)
 - 輸入 `npm install app-inspector -g`
 - 輸入 `macaca doctor`

連接手機

- ❖ 手機USB偵錯模式打開
- ❖ 確認手機ID(PC開啟命令提示字元)
 - 輸入 adb devices



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取得APP元素定位

- ❖ 執行App Inspector(開啟命令提示字元)
 - 輸入 `app-inspector -u <YOUR-DEVICE-ID>` (例如 `app-inspector -u [REDACTED]`)

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(contained account / credential data)

執行測試案例

- ❖ 啟動Appium(以系統管理員身分開啟命令提示字元), 如下圖
 - 輸入 appium
- ❖ 撰寫並執行測試腳本

Administrator: Command Prompt - appium

```
Microsoft Windows [Version 10.0.15063]
```

```
(c) 2017 Microsoft Corporation. All rights reserved.
```

```
C:\WINDOWS\system32>appium
```

```
[Appium] Welcome to Appium v1.7.1
```

```
[Appium] Appium REST http interface listener started on 0.0.0.0:4723
```

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screenshot removed
(contained account / credential data)

常用關鍵字

- ❖ Open Application
- ❖ Click Element
- ❖ Wait Until Element Is Visible
- ❖ Swipe
- ❖ Swipe by percent
- ❖ Shake
- ❖ Close All Applications

參考文件

❖ APP範本

➤ \\ [redacted]

❖ 如果Macaca無法安裝，改用Android內建viewer

➤ 打開 cmd:

- cd C:\Users\<user_name>\AppData\Local\Android\Sdk\tools\bin
- uiautomatorviewer.bat

❖ Android XPath定位方式

➤ <http://www.software-testing-tutorials-automation.com/2015/10/ui-automator-viewer-get-android-app.html>

❖ 下載 [redacted] APK檔

➤ [http:// \[redacted\] /apk/download.html](http:// [redacted] /apk/download.html)

APP 自動化測試分配

- ❖ **1/19** 建立Android自動化測試環境，並試跑快3範本(████.robot)，確認可以正常執行
- ❖ (TBD) 撰寫APP遊戲測試腳本
 - 時時彩、11選5： <colleague>
 - 快3、快樂彩： <colleague>
 - PC蛋蛋、六合彩： <colleague>
 - 排列三/五、福彩3D、平台： <colleague>